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Project Title:	Ground Based Measurement of Bone Loss in Astronauts Using AMPDXA GCS		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Technology Development Team		
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Comments:			
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Contact Monitor:		**Contact Phone:**	
Contact Email:			
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Key Personnel Changes/Previous PI:			
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Task Description:	The Johns Hopkins University Applied Physics Laboratory and the School of Medicine were building a Ground-based Clinical System using Advanced Multiple Projection Dual-energy X-ray Absorptiometry (AMPDXA) principles to go beyond bone densitometry to measure the structural mechanics of bones and muscles. Exposure to microgravity during long-duration space flight decreases bone and muscle mass but the implications for injury differ from similar losses on earth and cannot be discerned by densitometry. The goal of the Advanced Multiple Projection Dual Energy X-ray Absorptiometry Ground-Based Clinical System (AMPDXA-GCS) was to produce an operational instrument that could be transferred to the NASA-Johnson Spaceflight center for use in the pre-and post-flight bone mineral density (BMD) and bone structure measurements on astronauts. The original aims of the project include: 1. Design, fabrication, and testing of AMPDXA-GCS. 2. Validate the utility, radiation safety, and method of operation of the AMPDXA-GCS configuration. 3. Address key technical challenges necessary to migrate the AMPDXA-GCS from ground-based studies to spaceflight. 4. Transfer AMPDXA-GCS technology into commercial clinical uses. The AMPDXA-GCS will be able to detect 1% changes in bone mass and geometry and 2% changes in muscle mass and it will permit biomechanical assessments with similar accuracy. It will also decompose soft tissue into fat and muscle. Its accuracy and broad measurement capability make the AMPDXA-GCS superior to any conventional means for assessing bone health and strength. An imaging system with AMPDXA-GCS qualities in space would be an unrivaled tool for rapid, high-precision testing of the effects of countermeasures on the musculoskeletal system. Thus, AMPDXA-GCS technology is highly likely to result in new tests for countermeasures efficacy and it promises to make a significant contribution to NASA Critical Path Roadmap (CPR) risk reduction. AMPDXA-GCS monitoring techniques could also benefit millions of people suffering from osteoporosis and other bone disorders. Research and clinical medicine will both benefit from increased ability to more precisely monitor bone health and the efficacy of treatments. Plus, the design principles will allow the development of a compact less expensive version for limb imaging that could be conveniently used in a variety of remote settings such as rural clinics and nursing homes. The AMPDXA-GCS builds on the successes demonstrated in previous NSBRI funded collaborations between the Johns Hopkins University Applied Physics Laboratory (APL) and the Johns Hopkins University School of Medicine (SOM) which resulted in an AMPDXA Laboratory Test Bed (LTB) and a Human Test Bed (HTB). Since the AMPDXA-GSC will operate on slightly different principles (three X-ray beams at fixed angles) than the previous system design it was necessary to demonstrate the new concept using a test system-the AMPDXA-GCS Test Bed (ATB). The ATB was constructed and initial operating tests verified the new single scan with three fixed beams principles. The orginal 4-year project was terminated in its second year due to lack of future funding by the NSBRI. With the realization that because of the lack of funding the AMPDXA-GCS would not be built, the investigators turned their attention and remaining funding (with permission of the NSBRI) towards demonstration of the AMPDXA-GCS principles on a commercial DEXA scanner with the goal of commercializing the new technology invented under NSBRI funding. To this end the JHU team began working with the latest Hologic DEXA scanner and was able to modify text based scan protocols available in the system to allow multiple projection imaging. Multiple projection images were obtained from a sample bone at a constant distance by properly rotating the C arm and translating the patient table. We were then able to convert the Hologic data files into x ray attenuation files using a visual basic program. Software operations on these files resulted in bone mineral decomposition files. Several scans were made on phantoms and various bone samples such as a pig femur. At this point we were able to use our previous software to calculate bone cross sectional properties and compare the results to previous scans of the same bone using CT. A good comparison resulted between the Hologic derived properties, CT scan (used as reference), and our previous AMPDXA scanner (HTB). This proved that a commercial system could be modified to provide improved bone analysis embracing AMPDXA-GCS principles. Because of lack of source code, the scanning protocols developed were slow and would need to be improved before human testing could begin. The JHU Team was in discussions with Hologic about source code and funding future efforts when the project terminated due to lack of funding.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The AMPDXA project has many implications for future research and development. The AMPDXA-GCS, as described above, has direct application to risk reduction in NASA's Critical Research Path. The AMPDXA-GCS is capable of real-time monitoring of bone and muscle loss, at any anatomical location, with extremely high precision. Since the results are patient-specific and not tied to volumetric averages and statistical norms, the AMPDXA-GCS is a useful tool for monitoring the effectiveness of countermeasures as well as determining the risk of fracture under various loading conditions and activity scenarios. The AMPDXA-GCS also appears to be a natural adjunct to earthbound research on the effect of aging and disuse on bone integrity. It could also be used as a routine screening tool for osteoporosis and as a monitoring instrument for osteoporosis drug therapy. Since the AMPDXA-GCS measures structure to high precision and does an excellent job at imaging not only bone but also manmade prosthetic implants, the AMPDXA-GCS could be used by orthopedic surgeons to study the life progression of implants (e.g., loosening, bone regression, etc.). Given its low radiation dose and ability to automatically determine dimensions from the collected radiographic data, the AMPDXA-GCS may prove valuable in other orthopedic applications as well. For example, it is possible to predict stress fractures from BMD measurements. Stress fractures in military recruits range from one to seven percent, depending upon gender and branch of service. An untold number of athletes and other individuals also suffer stress fractures annually as a result of intense repetitive exercise. Such exercise or other similar activity can tire muscles so that they are no longer able to absorb the shock from these activities. As a result, tiny hairline cracks known as stress fractures develop in the bone. The best treatment for stress fractures is rest for six to eight weeks. Such rest periods would prevent a professional athlete or a soldier from performing his/her job. Thus, the ability to identify persons susceptible to stress fractures would enable a different training regimen for these persons in order to

	prevent the onset of a stress fracture. Commercially, if cost projections can be realized, there is a significant market for population screening and treatment monitoring of osteoporosis. Based on the AMPDXA-GCS, an easy-to-use, relatively compact instrument could be developed for a small clinic or doctor practice. The unit could be either horizontal like the GCS or vertical (patient stands). Concepts for a vertical unit have been developed. The market for such a commercial AMPDXA could range from several hundred to several thousands of units, depending upon final cost. If the unit could be structured to be lightweight and relatively transportable, it would have important implications for use as a tool in nursing homes, clinics, and doctor's offices. Several key members of the NSBRI's Bone Research Team have already indicated that a precision AMPDXA would greatly assist their work in bone loss studies, especially in bed rest studies. AMPDXA commercialization efforts are being pursued by both the APL Technology Transfer Office and the Technology Transfer Office at the Johns Hopkins Medical Institutions. Commercialization of the AMPDXA software for use with conventional DXA data may provide the stepping-stone for the commercialization of the AMPDXA hardware.
Task Progress:	A bench-top version (ATB) of the AMPDXA-GCS system has been fabricated to test out the physical principals of the new AMPDXA system. The ATB system is operational and images of cadaver bones and phantoms have been made. A new higher accurate phantom for new geometry has been deigned and is in use. The x-ray tube produces a beam that is parallel to the plane of the table and detected by 3 detectors positioned at zero, minus 15, and plus 15 degrees apart. Lead collimators are used to limit the beam to just the area of the detectors. A linear motion stage is used to move a test sample through the beam which simulates the motion of the C-arm relative to the patient in the final AMPDXA-GCS design. The bench-top also allows adjustment of tube to sample ("patient") distance in order to optimize the final design. The tube to detector distance can also be varied on the bench-top as part of this optimization. The x ray beam intensity was monitored by a scintillating detector diode located near the x ray source. This detector diode is referred to as the beam monitor. Beam intensity information allows the system to monitor patient dose and can be used to correct detector data for intensity variation and drift. It was decided to make the beam monitor a separate USB device. The microcomputer and analog board designs were merged onto one board and connectors for the scintillation detector were added. Although the electronics of this USB device are not placed in an aluminum enclosure, the beam monitor is completely enclosed by a black Kapton enclosure. The beam monitor and respective electronics have been manufactured and fully tested. With the hardware and software components of the testbed integrated, the next phase of the project was to evaluate the detector performance. A series of tests were developed and executed that measured the linearity response, dynamic range, and signal to noise ratio (SNR) properties of the detector. While the dynamic range and linearity tests were straight forward, the SNR tests and analysis were much more complex. The reason for this complexity involves patient size variability. Theoretically, the SNR is a function of the difference between the two kilovoltages used. Because attenuation is always much greater at the low kVp for thicker patients, the optimal combination of kVp and beam filter is not readily apparent for thicker patients. In order to quantify the detector SNR a stepped phantom was built using various thicknesses of plastic and aluminum. Our optimization experiments used four different thicknesses of plastic to simulate different patient thicknesses and we examined the SNR produced in a nominal thickness of aluminum The upper kVp was fixed at 140 kVp and the lower was varied from 70 to 110 kVp in 5 kVp increments. We concluded that with the current detector configuration a low kVp of 110 would provide a reasonable SNR with thicker patients.
Bibliography Type:	Description: (Last Updated: 12/16/2008)
Patents	US Patent 7,186,022. Patent, March 2007. Mar-2007 Charles HK, Bech TJ. "X-ray Source and Method for More Efficiently Producing Selectable X-ray Frequencies."
Patents	US Patent 7,203,956. Patent, April 2007. Apr-2007 Charles HK, Beck TJ, Magee TC, Feldmesser HS. "Method and Apparatus for Multiple-Projection, Dual Energy X-ray Absorptiometry Scanning."